ESAME 2

Come già calcolato nell'es. 7.1 del Kennett:

→ modi di disporre N bosoni in M orbitali:

$$\frac{(N+M-1)!}{N!(M-1)!}$$

→ modi di disporre N fermioni in M orbitali:

$$\frac{M!}{N!(M-N)!}$$

Quindi se ho n_s bosoni in g_s stati con energia E_s

e un numero arbitrario di particelle per stato:

$$\Omega_s = \frac{(g_s + n_s - 1)!}{n_s!\,(g_s - 1)!} \qquad [\log N! \simeq N \log N - N]$$

$$\frac{S}{k} = \log(\Omega_{TOT}) = \log\left(\prod_s \Omega_s\right) = \sum_s \log(\Omega_s)$$

$$\log(\Omega_s) \simeq (g_s + n_s - 1)\log(g_s + n_s - 1) - \cancel{(g_s + n_s - 1)}$$
$$- n_s \log n_s + \cancel{n_s} - (g_s - 1)\log(g_s - 1) + \cancel{(g_s - 1)}$$

Sia $\bar{n}_s = \frac{n_s}{g_s}$

$$= [g_s (1 + \bar{n}_s) - 1] \log[g_s(1 + \bar{n}_s) - 1] - g_s \bar{n}_s \log(g_s \bar{n}_s) - (g_s - 1)\log(\dots$$

AFFINCHÉ PORTI IL RISULTATO TRASCURO GLI 1:

$$\begin{cases} g_s(1 + \bar{n}_s) - 1 \simeq \cancel{g_s + \bar{n}_s} = \alpha + \beta \\ g_s \bar{n}_s \equiv \alpha \\ g_s - 1 \simeq g_s \equiv \beta \end{cases}$$

$$\log \Omega_s \simeq \underbrace{g_s \log(\alpha + \beta)} + \underbrace{g_s \bar{n}_s \log(\alpha + \beta)} - \underbrace{g_s \bar{n}_s \log(\alpha)} - \underbrace{g_s \log(\beta)}$$

$$= g_s \bar{n}_s \log\left(\frac{\alpha + \beta}{\alpha}\right) + g_s \cancel{\bar{n}_s} \log\left(\frac{\alpha + \beta}{\beta}\right)$$

$$\begin{cases} \frac{\alpha + \beta}{\alpha} = 1 + \frac{\beta}{\alpha} \simeq 1 + \frac{1}{\bar{n}_s} = \frac{\bar{n}_s + 1}{\bar{n}_s} \\ \frac{\alpha + \beta}{\beta} = 1 + \frac{\alpha}{\beta} \simeq 1 + \bar{n}_s \end{cases}$$

$$= \bar{n}_s g_s \left[\log(\bar{n}_s + 1) - \log(\bar{n}_s)\right] + g_s \cancel{\bar{n}_s} \log(1 + \bar{n}_s)$$

$$\cancel{= g_s [(1 + \bar{n}_s)\log(1 + \bar{n}_s)}$$

$$= g_s \left[(1+\overline{n}_s) \log(1+\overline{n}_s) - \overline{n}_s \log \overline{n}_s \right]$$

$$\Rightarrow \boxed{S = k_B \sum_s' g_s \left[(1+\overline{n}_s) \log(1+\overline{n}_s) - \overline{n}_s \log \overline{n}_s \right]}$$

Massimizzo S coi vincoli: $\sum' n_s \varepsilon_s = U$ e $\sum' n_s = N$, N, U = costanti

$$\mathcal{L} = -k_B \sum_s' g_s \left[(1+\overline{n}_s) \log(1+\overline{n}_s) - \overline{n}_s \log \overline{n}_s \right] - a \left[\sum' g_s \overline{n}_s \varepsilon_s - U \right] - b \left[\sum' g_s \overline{n}_s - N \right]$$

Impongo: $\frac{\partial \mathcal{L}}{\partial \overline{n}_s} = 0$ Massimizzo senza k_B come sempre

$$\cancel{\frac{\partial \mathcal{L}}{\partial \overline{n}_s} = + k_B \left[g_s \log(1+\overline{n}_s) + g_s \right] + a g_s \varepsilon_s + b g_s = 0}$$

$$\cancel{\log(1+\overline{n}_s) = -1 - a\varepsilon_s - b}$$

$$\cancel{\Rightarrow \overline{n}_s =}$$

$$\frac{\partial \mathcal{L}}{\partial \overline{n}_s} = -g_s \left[\log(1+\overline{n}_s) \cancel{+1} - \log \overline{n}_s \cancel{-1} \right] - a g_s \varepsilon_s - b g_s = 0$$

$$\Rightarrow \log\left(\frac{1+\overline{n}_s}{\overline{n}_s}\right) = -a\varepsilon_s - b \qquad e$$

$$\frac{1+x}{x} = \alpha \qquad 1+x = \alpha x \qquad x(1-\alpha) = -1 \qquad x = \frac{1}{\alpha - 1}$$

$$\Rightarrow \overline{n}_s = \frac{1}{e^{-a\varepsilon_s - b} - 1}$$

Sia $\alpha = -b$, $\beta = -a$

$$\Rightarrow \overline{n}_s = \frac{1}{e^{\alpha + \beta \varepsilon_s} - 1}$$

Es 2

→ Per particelle classiche distinguibili non ho limitazioni sul numero di particelle in uno stato ma, essendo distinguibili considero ~~no~~ ripetizione: ho

VEDI APPUNTI MECCANICA STATISTICA